National University of Computer and Emerging Sciences

Information Security (CS3002)

Date: Monday, Sep 21st, 2026

Instructor (s)

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Sessional-I Exam

Total Time: 1 Hours

Total Marks: 15

Total Questions: 05

Semester: FALL-2026

Campus: Karachi

Dept: BCS-7,BSE-7,BCY-5

Student Name	Roll No	Section	Student Signature

Instructions:

- Each question is set to have a specific answer. Unrelated longer answers will not be read nor graded nor responded to during review.

CLO # 1: Explain key concepts of information security such as design principles, cryptography, risk management, and ethics

Question 1 [2 Marks]

The Avalanche Effect refers to a property of cryptographic algorithms in which a small change in the input (plaintext or key) — even a single bit — produces a significant, seemingly random change in the output (ciphertext), typically causing about half of the output bits to flip. Identify the cryptographic properties/functions through which the Avalanche Effect is achieved, and justify your answer.

Question 2 [2 Marks]

Consider a student information system (SIS) in which students provide a university student number (USN) and a card for account access. Give one example each of a confidentiality, integrity, and availability requirement associated with this system, and in each case, indicate the degree of importance of the requirement (e.g., high, moderate, low), with justification.

CLO # 3: Analyze and apply various security and risk management tools for achieving information security and privacy.

Question#3: [5 Marks]

RSA (public-key cryptography) and AES (symmetric-key cryptography) are both widely used, yet real-world systems (e.g., TLS/HTTPS) almost always use RSA (or another public-key algorithm) only to exchange a key, and then switch to AES for the actual bulk data encryption.

(a) Explain the underlying mathematical/computational reason why RSA is significantly slower than AES for encrypting large volumes of data. (2 Marks)

(b) Justify, with reasoning, why using RSA for key exchange and AES for bulk encryption (a hybrid cryptosystem) is a better design decision than using either algorithm alone for the entire communication. (2 Marks)

(c) If a digital signature (using RSA and a hash function) is added to this hybrid scheme, what additional security service does it provide that encryption alone does not? (1 Mark)

CLO # 4: Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security

Question#4: [2 Marks]

Read the case study below. Identify which component(s) of the CIA Triad has been breached, explain how the breach occurred, and describe the potential impact of the breach on the organization and its stakeholders.

Case Study:

A hospital's Electronic Health Record (EHR) system experienced a security incident. A disgruntled IT contractor, who still had active login credentials from a previous project, accessed the system and altered dosage information on several patients' prescription records before his access was noticed and revoked three days later. During those three days, doctors continued prescribing medication based on the tampered records.

Question#5: [4 marks]

Draw a compact free-hand labeled diagram to explain the following scenarios:

i) A bank sends fund-transfer instructions to a partner server and, without encrypting the message, wants the receiver to be able to verify that the amount and account details were not altered in transit and genuinely came from someone sharing the bank's secret key — describe, with a labeled diagram, how this can be achieved.

ii) Alice receives a message M from Bob, along with a signature S that Bob generated using his private key. Alice does not have Bob's private key and cannot access it, but she does have Bob's public key. Describe, with the help of a labeled diagram, the complete process Alice would follow to verify that the message genuinely came from Bob and was not altered in transit — without needing any shared secret key between herself and Bob.